

KRISHNA KANT GARHE

Electronics Engineer | Embedded Systems, Hardware & PCB Design | Applied Software Development

Durg, Chhattisgarh | +91 76949 83068 | krishnakantgarhe@gmail.com | Open to Relocation Across India

PROFESSIONAL SUMMARY

Electronics & Telecommunication Engineering graduate (BIT Durg) with hands-on experience spanning analog and mixed-signal circuit design, PCB design, embedded systems, and applied software development. Skilled in schematic design, analog front-end design, and R-2R DAC/ADC circuit design using KiCad and Proteus, alongside microcontroller programming (ESP32, Arduino, Raspberry Pi Pico) and protocol-level interfacing (UART, SPI, I²C). Proficient in C, C++, and Python with strong object-oriented programming knowledge, version-controlled development using Git/GitHub, and applied machine learning (SVM, Random Forest, CNN) for sensor data classification. Delivered several end-to-end hardware and embedded products, including a NIR-based microplastic analyzer achieving 95%+ classification accuracy and a custom analog front-end for a portable digital oscilloscope. Excellent analytical and problem-solving skills, with proven ability to debug across hardware and software layers and collaborate effectively in cross-functional engineering teams. Seeking roles in Hardware Engineering, Embedded Systems, Firmware, or Applied Software Development.

TECHNICAL SKILLS

Analog & Mixed-Signal Design: Analog Front-End Design, R-2R Ladder DAC/ADC Circuit Design, PLL/Op-Amp Based Signal Conditioning (project-level), Overvoltage Protection, AC/DC Coupling

PCB & Circuit Design: KiCad, Proteus, Schematic Design, PCB Design, Circuit Simulation, Hardware Prototyping & Validation

Programming & Version Control: C, C++, Python, Embedded C, Object-Oriented Programming, Data Structures (applied via project work), Git, GitHub

Embedded Systems: ESP32, Arduino Uno, Raspberry Pi Pico (RP2040), Microcontroller Programming, Sensor Interfacing, Board-Level Firmware Development

Communication Protocols: UART, SPI, I²C, Wi-Fi (ESP32/ESP8266 connectivity), Serial Communication, Data Acquisition

Machine Learning: SVM, Random Forest, CNN, Data Preprocessing, NIR Spectral Analysis, Signal Processing, Data Analysis, Model Benchmarking

Debugging & Problem Solving: Hardware-Software Fault Diagnosis, Protocol-Level Debugging (I²C/SPI/UART), Circuit-Level Troubleshooting, Analytical Problem Solving

Platforms & Tools: Linux (Ubuntu, dev environment), VS Code, Arduino IDE, Blynk 2.0

Core Strengths: Excellent Analytical & Problem-Solving Skills, Cross-Functional Team Collaboration, Verbal & Written Communication, Efficiency-Driven Design

KEY PROJECTS

NIR-Based Microplastic Analyzer — Flagship, Final Year

Jul 2025 – Jan 2026

ESP32 · AS7263 NIR Sensor · SVM · Random Forest · CNN · Python · I²C · Wi-Fi

- Designed a portable embedded sensing device using ESP32 and the AS7263 six-channel NIR sensor to detect and classify 5 microplastic polymer types (PET, PE, PP, PS, PVC) via I²C data acquisition and wireless transmission.
- Benchmarked SVM, Random Forest, and CNN classifiers on NIR spectral data, achieving 95%+ classification accuracy after feature engineering — delivering a field-deployable, real-time alternative to lab analysis.
- Built and integrated an end-to-end embedded data pipeline: sensor acquisition → real-time preprocessing → on-device ML inference → Wi-Fi transmission for remote environmental monitoring.

Arbitrary Waveform Generator — Raspberry Pi Pico

Jan 2024 – May 2024

RP2040 · Programmable I/O (PIO) · R-2R DAC · MicroPython

- Built a 9-waveform real-time signal generator (sine, square, triangle, Gaussian, sawtooth, and more) using the RP2040 PIO subsystem, generating frequencies up to 10 MHz via a custom R-2R ladder DAC.
- Designed an LCD-based real-time control interface for waveform selection and parameter tuning, demonstrating complete embedded hardware-firmware co-design.

PicoScope — DIY Portable Digital Oscilloscope

Jul 2023 – Dec 2023

Raspberry Pi Pico · Scopy App · Analog Front-End Design

- Developed a portable oscilloscope and logic analyzer with a custom analog front-end (AC/DC coupling, overvoltage protection, ±25V range) for real-time waveform visualization via the Scopy smartphone app.
- Implemented on-device I²C, SPI, and UART protocol decoding, providing a low-cost, practical embedded systems debugging tool.

IoT-Based Automatic Medicine Dispenser

Jan 2023 – May 2023

Arduino Uno · ESP8266 Wi-Fi · Blynk 2.0 · Servo Motors

- Built an IoT-enabled embedded system using Arduino Uno, ESP8266, and Blynk 2.0 for cloud-connected, scheduled dosing with real-time push notifications, supporting medication adherence for elderly patients.

- Integrated a servo-controlled delivery mechanism and LED status indicators for contactless operation, covering complete IoT system design from embedded hardware through cloud connectivity.

WORK EXPERIENCE

Onsite Engineer (Trainee)

Feb 2026 – May 2026

Vaishnavas Energy Pvt. Ltd. (VEPL), Bhilai — Clean Energy & Battery Technology | Lithium Battery, BMS, EV & BESS Systems

- Performed lithium battery pack assembly, BMS wiring, and electrical systems integration for EV, BESS, and agricultural energy storage applications; conducted quality and functional testing to ensure safety and performance compliance.
- Gained industrial-scale exposure to clean-energy product development, including electric mobility solutions and renewable energy storage systems.

Summer Engineering Intern

Jun 2025 – Jul 2025

Chhattisgarh State Power Generation Company Limited (CSPGCL), Raipur — Government Power Utility

- Gained hands-on exposure to SCADA systems for real-time monitoring and automated control of thermal, hydro, and renewable power generation infrastructure at state-utility scale.

Aerospace & Drone Systems Training Program

Feb 2026 – Mar 2026

India Space Lab (ISL) — Remote Training | Drone Technology, CanSat/CubeSat, Remote Sensing & GIS

- Completed structured, project-based training in advanced drone systems, CanSat/CubeSat satellite engineering, rocketry, and remote sensing & GIS.

Documentation & Coordination Intern — Solar Subsidy Processing

Mar 2022 – Apr 2022

Unique Associates, Raipur — Private Contractor under CREDA, Government of India | Surya Sujala Yojana

- Supported rural solar panel installation (3W & 5W) and digitized farmer subsidy application forms under the Surya Sujala Yojana scheme, improving processing accuracy and administrative efficiency.

Teaching & Mentoring Experience

2022 – Present

Mentee to Mentor Academy, Bhilai (Jun 2025–Present) | BrainStormers' Academy, Bhilai (2022–2026)

- Delivered curriculum-aligned instruction for Class 10–12 and NEET Dropper batches; designed assessments and applied 1:1 mentorship to close learning gaps across Physics, Chemistry, Mathematics, and Biology.
- Prepared JEE/NEET aspirants through structured doubt-clearing and strategic problem-solving, consistently achieving measurable improvement in student outcomes.

EDUCATION

Bhilai Institute of Technology (BIT), Durg, Chhattisgarh

B.Tech — Electronics & Telecommunication Engineering | CGPA: 7.0 / 10.0

Relevant Coursework: Embedded Systems | Microprocessors & Microcontrollers | Digital Electronics | Analog Electronics | Control Systems | Signals & Systems | Communication Engineering | Machine Learning

CERTIFICATIONS

- C & C++ Programming — Spoken Tutorial Project, IIT Bombay
- Advanced Drone Technology & Space Systems — India Space Lab, ISL (2026)
- Electronics & PCB Design Crash Course — André LaMothe, Udemy (In Progress)

ACHIEVEMENTS

- Developed a NIR-based microplastic analyzer achieving 95%+ ML classification accuracy — a field-deployable environmental sensing solution.
- Advisor & Event Manager, Astro Club, BIT Durg (2024).
- NCC Cadet — developed leadership and discipline through structured training and official camps.
- NSS Volunteer — recognised for consistent contribution to community service and social awareness initiatives.

CORE COMPETENCIES

Technical: Analog & Mixed-Signal Circuit Design | PCB Design | Hardware Prototyping | Circuit Simulation | Embedded Firmware Development | Python Programming | Applied Machine Learning | Git/GitHub Version Control | Debugging & Troubleshooting | Sensor Integration | Data Acquisition | Signal Processing | Battery Management Systems (BMS) | Power Electronics

Languages: English (Professional Working Proficiency) | Hindi (Native)